

Should every baby's DNA be sequenced?

The genomic generation is on its way

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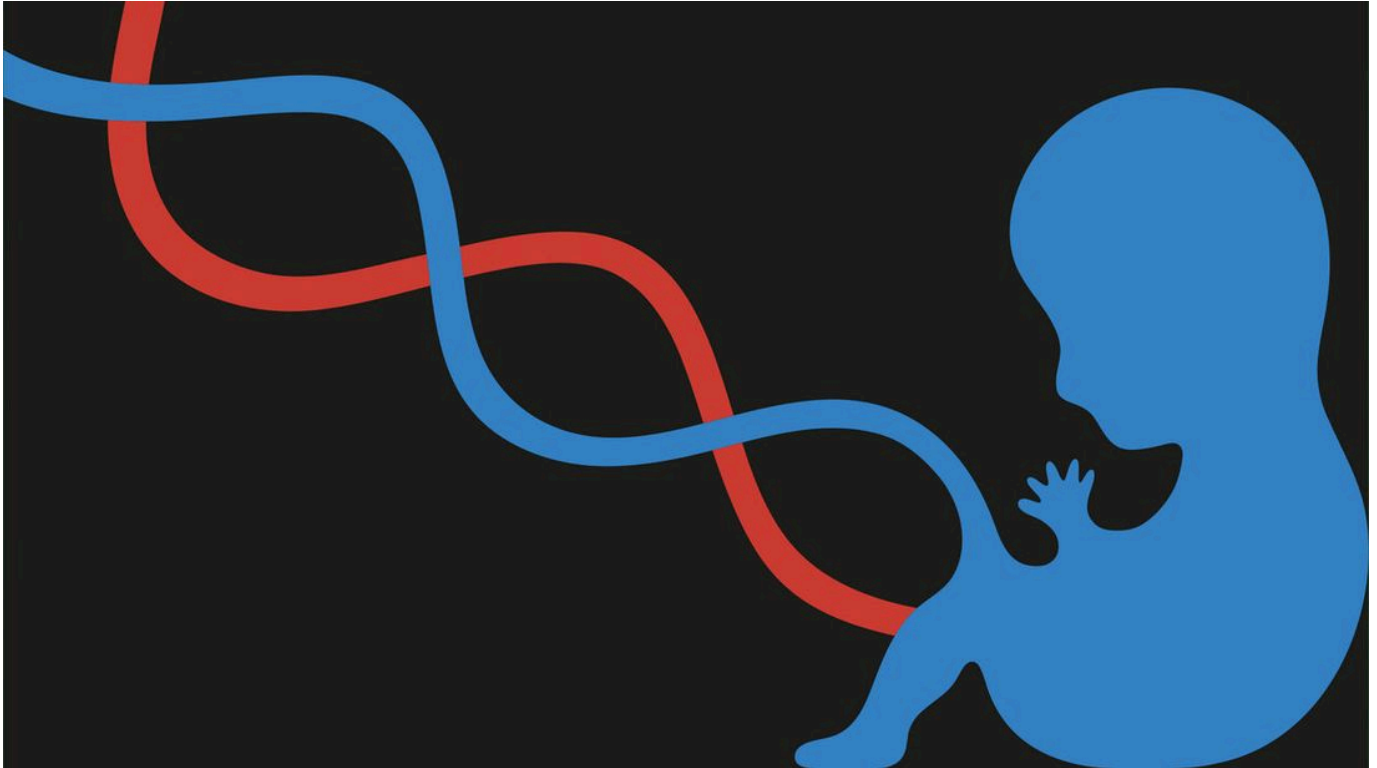


Illustration: Andreas Samuelsson

SHORTLY AFTER Freddie was born in April 2025, he was diagnosed with retinoblastoma, a rare form of eye cancer. It was spotted early and treated promptly, which means he has a much better chance of growing up with normal vision. That quick diagnosis was no accident. Freddie is part of the Generation Study, an English programme which plans to sequence the complete genetic code of 100,000 babies. The idea is both to screen them for genetic diseases and to use the collected genomic data to boost medical research in future.

Similar trials are under way in America, Australia and parts of Europe, to test whether such screening should be offered for all newborns. Supporters hope to transform the diagnosis and treatment of rare diseases. Critics fret that genomic sequencing may cause needless worry, and require parents to hand over sensitive data about their children.

Many countries screen infants for diseases already, usually by taking a blood sample and looking for markers of conditions such as sickle-cell anaemia or cystic fibrosis. The most comprehensive screens, such as Italy's, look for markers of around 50 diseases. Analysing DNA rather than blood can increase that number dramatically, as well as flagging mutations that are not causing problems yet, but might in future. The Generation Study is screening for around 200 conditions

that arise in childhood. In America the BabySeq2 project screened for around 1,000 genes, some linked to diseases that emerge in adulthood such as breast or ovarian cancer.

Catching more diseases earlier could help many more babies like Freddie. But there are worries to go with the enthusiasm. One is that genes are not always destiny. The average person carries scores of genetic variants that have been linked to diseases. But because of environmental factors, the influence of other genes or chance, most of those conditions never develop. Geneticists use a concept called “penetrance” to describe the proportion of people with a disease-causing mutation who actually go on to become ill. It can vary greatly for different conditions—and many estimates may be too high. Many disease-causing mutations are found by working backwards from people who are already ill, which risks inflating the numbers.

Take retinoblastoma. Early studies estimated that mutations in a gene called *RB1* had penetrance of more than 90%. But a paper posted online in December found that less than a third of adults with risky variants have ever had the cancer. “I think historically people have equated genetic information with certainty,” says Caroline Wright a geneticist at the University of Exeter, who conducted the study, “which is quite outdated now.”

Testing hundreds of genes with low or middling penetrance could create “patients-in-waiting” many of whom will never become ill. Such diagnostic purgatory is stressful enough that, in the case of cystic fibrosis, it has a name: “cystic fibrosis, screen positive inconclusive diagnosis” (CFSPID). Parents of CFSPID children often question the worth of getting the test results at all, says Anneke Lucassen, a geneticist at the University of Oxford.

Uncertainty is not the only worry. Some treatments can cause harm too. In the case of retinoblastoma a risky mutation triggers only eye tests. But for medullary thyroid cancer caused by mutations in the *RET* gene (where new research also suggests penetrance is much lower than had been thought), a common follow-up is to surgically remove the thyroid gland. That leaves patients dependent on artificial thyroid hormones for the rest of their lives.

Some of the issues have solutions. Results could be given for only genetic variants which give rise to treatable illnesses, and which are known to have high penetrance—though that might upset those who would prefer to know everything. Careful communication can improve understanding of a test’s limitations. Part of the point of the trials is to explore exactly these sorts of questions.

But screening is only half the story. A second goal of most programmes is to store the genomes for use in future research. Such databases can be very useful. Data from Britain’s 100,000 Genomes Project, which collected genomes from patients with rare diseases and their families, allowed researchers at Oxford to spot genetic variants that cause a neurodevelopmental disorder in children.

The data could also be used to predict drug side-effects and predict how well a patient will respond to treatment. It could also be used to flag people who might benefit from new treatments that did not exist when their genomes were first stored. “I’m completely confident that our ability to understand the genome will improve,” says Ewan Birney, director of the European Bioinformatics Institute. “AI is giving us a massive boost to that.”

On the other hand, a person’s genome is among the most personal and sensitive information there is. For some researchers and parents, the idea of storing it indefinitely is a sticking-point. There have been several leaks of genetic or medical information in recent years. In 2023, for instance, hackers stole the genetic and personal information of 6.9m customers of 23andMe, a beleaguered direct-to-consumer genetic-testing company.

The same advances in genetics that enable new treatments are also likely to enable new uses for genetic information outside medicine. In America some insurers can already use genetic-test results when setting premiums (in Australia and Britain this is mostly banned). Insights from DNA could be used for blackmail, for instance around questions of paternity.

Jan Friedman, a geneticist at the University of British Columbia, worries that many of the proposed screening programmes suffer from “mission creep”, with most requiring parents to sign up to both screening and the long-term storage of their child’s data for research. “You can’t take part in one without taking part in the other,” says Dr Lucassen of the University of Oxford, yet “they’re so different”.

Screening programmes are usually judged on whether the benefits to the patients outweigh the costs. Medical research, by contrast, is justified by benefits that are uncertain and often accrue to other people. Current genomic-screening trials for newborns are trying to do both at once, and with patients who are incapable of consenting to boot. The benefits could be enormous. But the ethics look tricky. ■

Correction: an earlier version of this article said Freddie had been treated with photo-activated chemotherapy. This has been amended.

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